

Question 1

a)

The **Pearson correlation coefficient (r)** measures the linear relationship between two variables. The formula is:

$$r = \frac{n \sum xy - \sum x \sum y}{\sqrt{\{n \sum x^2 - (\sum x)^2\} \{n \sum y^2 - (\sum y)^2\}}}$$

Where n=10.

Table 1: Regression (or Correlation) Summation Table for Bandwidth and Latency

X (Bandwidth)	Y (Latency)	X ²	Y ²	XY
12	50	144	2500	600
18	45	324	2025	810
10	60	100	3600	600
25	30	625	900	750
22	35	484	1225	770
30	25	900	625	750
28	28	784	784	784
20	40	400	1600	800
15	48	225	2304	720
26	27	676	729	702
$\sum X = 206$	$\sum Y = 388$	$\sum X^2 = 4662$	$\sum Y^2 = 16292$	$\sum XY = 7286$

Now,

$$r = \frac{10(7286) - (206)(388)}{\sqrt{\{10(4662) - (206)^2\} \{10(16292) - (388)^2\}}}$$
$$= -0.982$$

b)

The relationship between bandwidth and latency is negatively and strongly correlative.

Question 2

a)

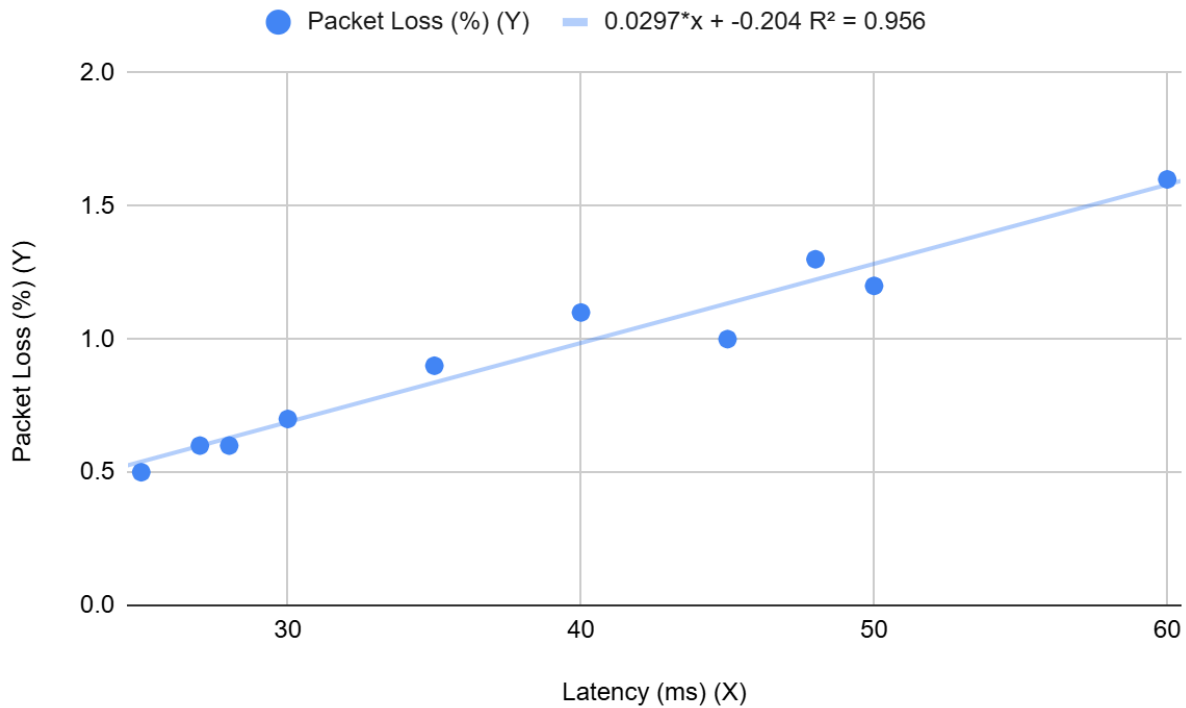


Figure 1: Scatter Plot

b)

The scatter plot reveals a strong positive linear relationship between latency and packet loss. As latency (ms) increases, packet loss (%) also tends to rise. The line of best fit is represented by the equation $y = 0.0297x - 0.204$ with $R^2=0.95$. This high value signifies that latency and packet loss are closely related; higher latency corresponds to a higher rate of packet loss.

Question 3

a)

The simple linear regression equation is $\hat{Y} = a + bX$

Where,

$$b = \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2}}$$

$$a = \hat{\beta} = \bar{Y} - b\bar{X}$$

Table 2: The dataset for Bandwidth (X) and Energy Consumption (Y)

X (Bandwidth)	Y (Energy Consumption)	X²	Y²	XY
12	2.8	144	7.84	33.6
18	3.2	324	10.24	57.6
10	2.5	100	6.25	25.0
25	4.0	625	16.00	100.0
22	3.7	484	13.69	81.4
30	4.5	900	20.25	135.0
28	4.3	784	18.49	120.4
20	3.4	400	11.56	68.0
15	3.0	225	9.00	45.0
26	4.1	676	16.81	106.6
$\sum X = 206$	$\sum Y = 35.5$	$\sum X^2 = 4662$	$\sum Y^2 = 130.13$	$\sum XY = 772.6$
$\bar{X} = 20.6$	$\bar{Y} = 3.55$			

$$\therefore b = \frac{10(772.6) - (206)(35.5)}{10(4662) - (206)^2}$$

$$= 0.098$$

$$\therefore a = 3.55 - 0.098 \times 20.6 = 1.5312$$

$$\begin{aligned} \text{The simple linear regression equation, } \hat{Y} &= a + bX \\ &= 1.5312 + 0.098X \end{aligned}$$

b)

Table 3: Regression Summation Table for Bandwidth (X) and Latency (Y)

X (Bandwidth, Mbps)	Y (Energy Consumption, W)	$\hat{Y} = 1.5312 + 0.098X$	$e^2 = (\hat{Y} - Y)^2$
12	2.8	2.7072	8.61184×10^{-3}
18	3.2	3.292	8.464×10^{-3}
10	2.5	2.5112	1.2544×10^{-4}
25	4.0	3.9812	3.5344×10^{-4}
22	3.7	3.6872	1.6384×10^{-4}
30	4.5	4.4712	8.2944×10^{-4}
28	4.3	4.2752	6.1504×10^{-4}
20	3.4	3.4912	8.31744×10^{-3}
15	3.0	3.0012	1.2544×10^{-4}
26	4.1	4.0792	4.3264×10^{-4}
$\sum X = 206$	$\sum Y = 35.5$		$\sum e^2 = 0.03357392$

$$\begin{aligned}
 \text{Now, } R^2 &= 1 - \frac{\sum e^2}{\sum y^2 - \frac{(\sum y)^2}{n}} \\
 &= 1 - \frac{0.03357392}{130.13 - \frac{35.5^2}{10}} \\
 &= 0.992 \\
 &= 99.2\%
 \end{aligned}$$

The fitted model has a good fit to the data and capable of explaining almost all data variation in the energy consumption (Y).

Interpretation of the Slope:

The slope b is 0.098 that means, if 1 Mbps bandwidth increased by 1 Mbps, on average the Energy Consumption of the device increased by 0.098 Watts.